Mobile system design interview cheat sheet

A short summary of mobiledesignrounds.com, for Android and iOS engineers. The method is my own. Companies differ.

THE FIVE STEPS

1. **Scope and constraints**: what the app must do, what it can skip, and the limits that shape the design.
2. **Data and contracts**: the data, the API, and the IDs that make a retry safe.
3. **Client structure**: the screen, the state holder, the repository, the local database and the network.
4. **Make the hard part work**: the send path, sync, caching or ordering.
5. **Break it, ship it, roll it back**: what can fail, how you would know, how the release goes out and how you roll it back.

45 MINUTES: A SUGGESTED SPLIT

- Scope and constraints: 4 min
- Data and contracts: 10 min
- Client structure: 8 min
- **Make the hard part work: 18 min**
- Break it, ship it, roll it back: 5 min

This is a suggested split, not a rule. Companies differ, but give the hard part the most time.

OFFLINE-FIRST CHECKLIST

- The local database is the source of truth. The screen reads only from it.
- Only the repository talks to the network. The screen updates when the data changes.
- Choose a write strategy for each kind of data: online only (payments), queued (analytics events), or lazy (a sent message).
- Lazy writes need a queue that survives a restart: WorkManager on Android, your own outbox on iOS.
- Every request carries an idempotency key (doing it twice has the same result as once), so a retry is safe.
- Sync with a cursor (a marker for the last item you got). Treat a push message as a hint, not as delivery.
- Settle conflicts with a server sequence number or version, not the device time.
- Limit the queue size: set a cap, drop the oldest item, count the drops.

FAILURE-MODE CHECKLIST

- The app is killed in the middle of a send or upload.
- A lost response followed by a retry (duplicate).
- A refresh that replaces a local change that is not sent yet.
- A slow old response that arrives after a new one.
- A device clock that is wrong.
- A missed push message or a closed socket.
- Old app versions that talk to a changed schema or protocol.
- A queue that grows for weeks while the phone is offline.

WHAT CHANGES AT SENIOR

- Lead the requirements with little help, and state your assumptions.
- Raise the key design points without being asked.
- Give reasons for your decisions and name the trade-offs.
- Name the failures: the app is killed, a response is lost, a refresh replaces an unsent change.

WHAT CHANGES AT STAFF

- Lead the conversation and narrow the problem on purpose.
- Cover rollout: flags, a staged release, a kill switch, what you watch and what causes a rollback.
- Cover metrics, privacy and old app versions.
- Say who owns what: shared modules, other teams, and build or adopt.
- Say clearly where iOS gives a weaker guarantee than Android.

READ MORE (ALL FREE)

The framework and 12 case studies: mobiledesignrounds.com/mobile-system-design/ | Start here (reading paths by level): /start/
Offline-first mobile system design: /guides/offline-first-hub/ | Staff-level mobile system design: /guides/staff-mobile-loop-guide/

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